

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA14

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO4: Examine intermediate code generation techniques to enhance code optimization (BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:40

Annexure A

CLASS TEST

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Q. No	Understanding of Concepts (25)	Explanation (25)	Application (20)	Organization (15)	Accuracy(10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

Annexure A

CLASS TEST

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	15%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps

1> Given Statement:

if $((a < b) \text{ and } ((c < d) \text{ or } (a > d)))$ then

$z = x + y * z$

else

$z = z + 1$

38/40

Sol:-

Three Address code for the statement:

L1:

if $a < b$ goto L2

goto L5

L2:

if $c < d$ goto L4

goto L3

L3:

if $a > d$ goto L4

goto L5

L4:

$t1 = y * z$

$t2 = x + t1$

$z = t2$

goto L6

L5:

$$t3 = z + 1$$

$$z = t3$$

L6:

2> Translate $(a < b)$ and $(c < d)$ or $(e < f)$ into three address ~~code~~ statements using back patching.

Sol:-

$$[(a < b) \text{ AND } (c < d)] \text{ OR } (e < f)$$

1. if $(a < b)$ goto 3

Backpatch Statement 1 to Statement 3

Backpatch Statement 2 to Statement 5.

Backpatch Statement

1. if $(a < b)$ goto 3

2. goto 5

3. if $(c < d)$ goto 7

4. goto 5

5. if $(e < f)$ goto 7

6. goto 8

7. True

8. False

3> Consider the arithmetic expression:-

$$A = (-A + B) * ((-A + B) * C)$$

Sol:-

Three Address Code for the Statement

$$t1 = -A$$

$$t2 = t1 + B$$

$$t3 = -A$$

$$t4 = t3 + B$$

$$t5 = t4 * C$$

$$t6 = t2 * t5$$

$$A = t6$$

(i) Quadruples:-

OP	arg 1	arg 2	result
-	A	-	t1
+	t1	B	t2
-	A	-	t3
+	t3	B	t4
*	t4	C	t5
*	t2	t5	t6
=	t6	-	A

(ii) Triples :-

Arg 1	Arg 2	result
A	-	t1
t1	B	t2
A	-	t3
t3	B	t4
t4	C	t5
t2	t5	t6
t6	-	A

4> Construct three address code using the following production for Boolean expression $(a > b) \text{ or } (c < d) \text{ or } (e < f)$

$B \rightarrow B1 \parallel B2$

$B \rightarrow B1 \text{ } \&\& \text{ } B2$

$B \rightarrow !.B1$

$B \rightarrow E1 \text{ relop } E2$

$B \rightarrow \text{true}$

$B \rightarrow \text{false}$

Sol:-

Three address code for the following statement:

L1:

if $a > b$ goto LTRUE

goto L2

L2:

if $c < d$ goto LTRUE

goto L3

L3:

if $e < f$ goto LTRUE

goto LFALSE

LTRUE:

// Boolean expression is true

goto LEND

LFALSE:

// Boolean expression is false

LEND: